# Solution: Unique Generalized Abbreviations

## Problem Statement

Given a word, write a function to generate all of its unique generalized abbreviations.

A generalized abbreviation of a word can be generated by replacing each substring of the word with the count of characters in the substring. Take the example of “ab” which has four substrings: “”, “a”, “b”, and “ab”. After replacing these substrings in the actual word by the count of characters, we get all the generalized abbreviations: “ab”, “1b”, “a1”, and “2”.

Note: All contiguous characters should be considered one substring, e.g., we can’t take “a” and “b” as substrings to get “11”; since “a” and “b” are contiguous, we should consider them together as one substring to get an abbreviation “2”.

**Example 1:**

```
Input: "BAT"
Output: "BAT", "BA1", "B1T", "B2", "1AT", "1A1", "2T", "3"
```

**Example 2:**

```
Input: "code"
Output: "code", "cod1", "co1e", "co2", "c1de", "c1d1", "c2e", "c3",
"1ode", "1od1", "1o1e", "1o2", "2de", "2d1", "3e", "4"
```

## Solution

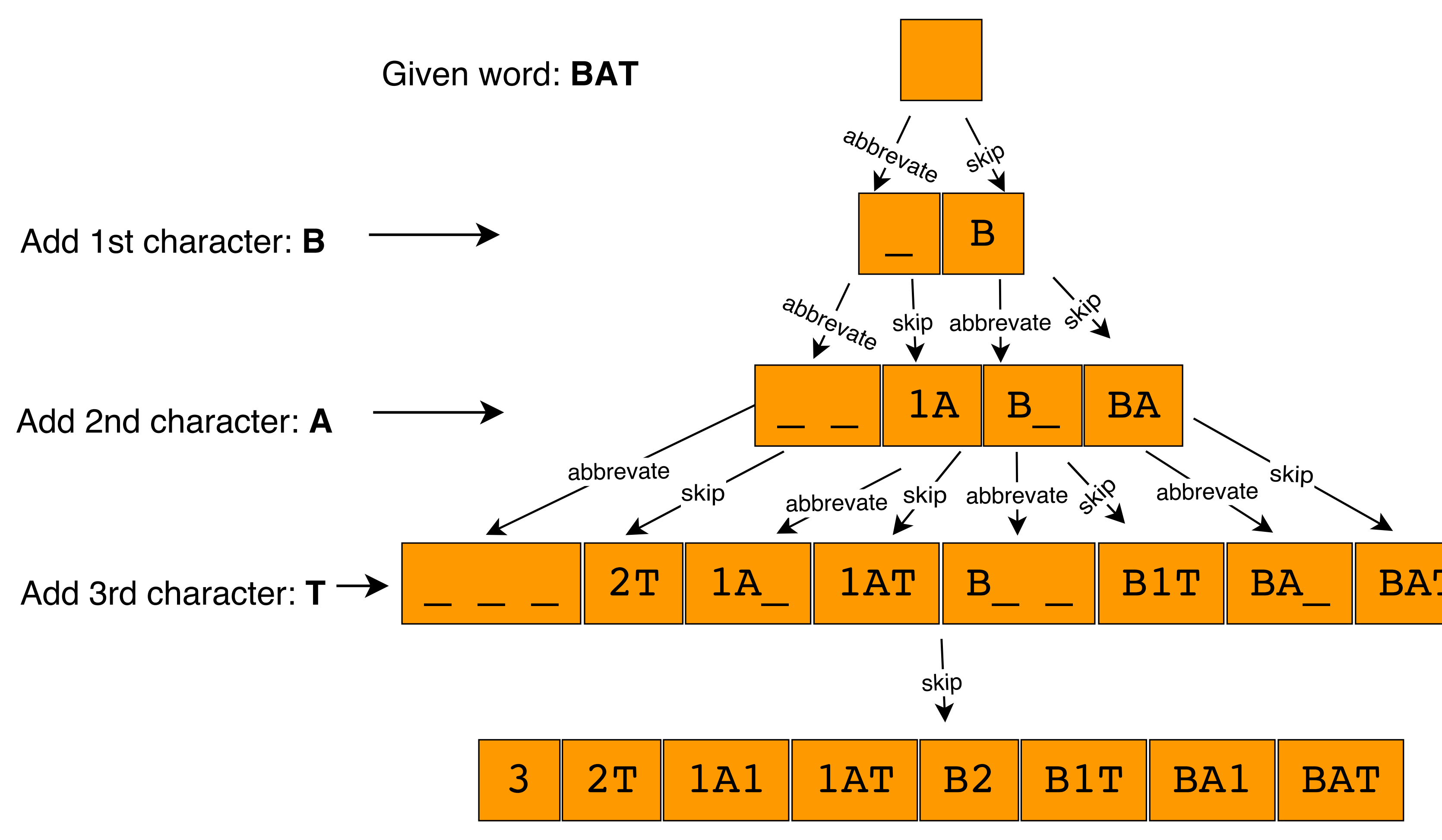
This problem follows the `Subsets` pattern and can be mapped to `Balanced Parentheses`. We can follow a similar BFS approach.

Let’s take Example-1 mentioned above to generate all unique generalized abbreviations. Following a BFS approach, we will abbreviate one character at a time. At each step, we have two options:

1. Abbreviate the current character, or
2. Add the current character to the output and skip the abbreviation.

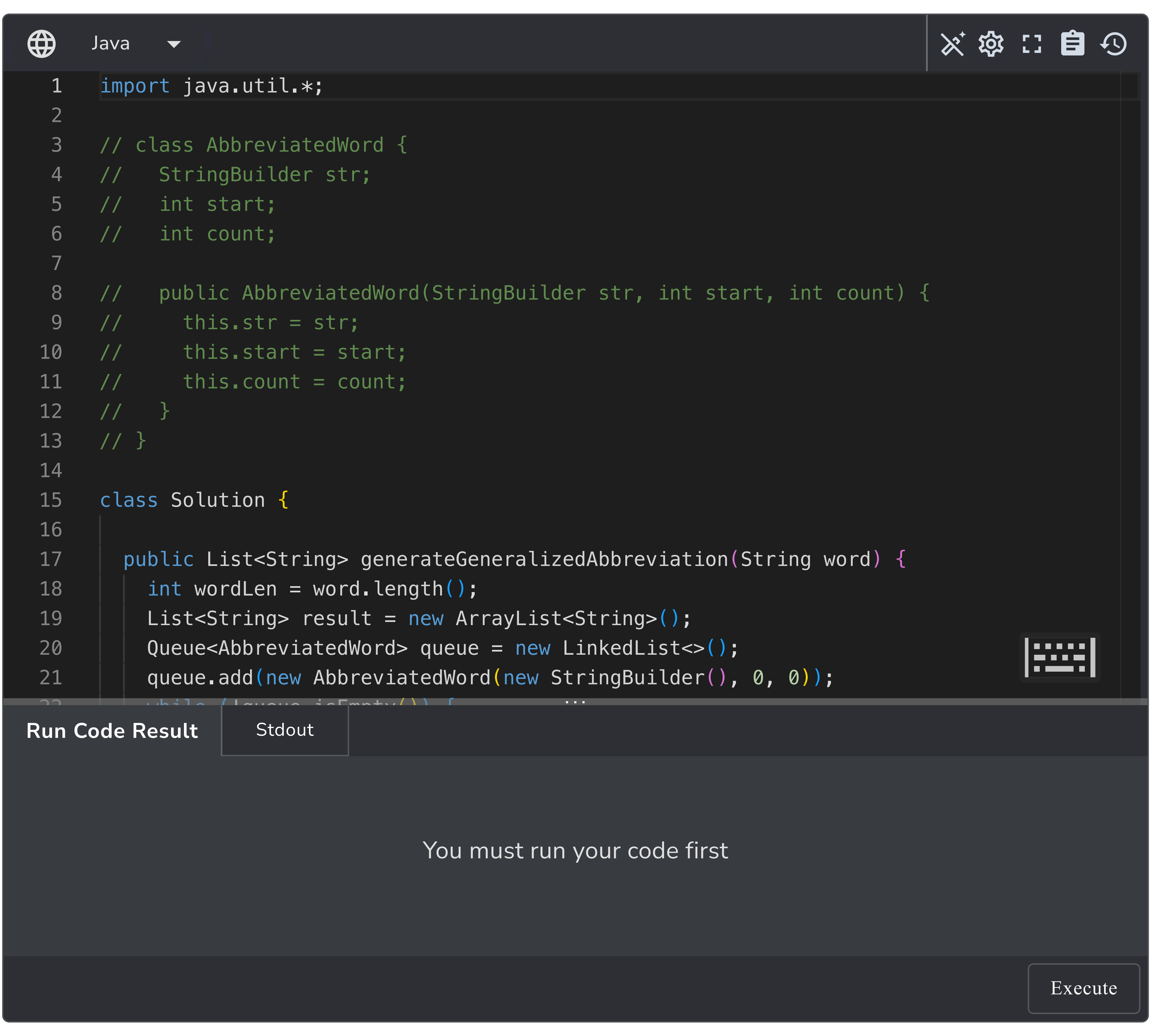
Following these two rules, let’s abbreviate `BAT`:

1. Start with an empty word: “”
2. At every step, we will take all the combinations from the previous step and apply the two abbreviation rules to the next character.
3. Take the empty word from the previous step and add the first character to it. We can either abbreviate the character or add it (by skipping abbreviation). This gives us two new words: `_`, `B`.
4. In the next iteration, let’s add the second character. Applying the two rules on `_` will give us `_ _` and `1A`. Applying the above rules to the other combination `B` gives us `B_` and `BA`.
5. The next iteration will give us: `_ _ _`, `2T`, `1A_`, `1AT`, `B _ _`, `B1T`, `BA_`, `BAT`
6. The final iteration will give us: `3`, `2T`, `1A1`, `1AT`, `B2`, `B1T`, `BA1`, `BAT` Here is the visual representation of this algorithm:



## Code

Here is what our algorithm will look like:



```java
import java.util.*;

// class AbbreviatedWord {
//   StringBuilder str;
//   int start;
//   int count;

//   public AbbreviatedWord(StringBuilder str, int start, int count) {
//     this.str = str;
//     this.start = start;
//     this.count = count;
//   }
// }

class Solution {

  public List<String> generateGeneralizedAbbreviation(String word) {
    int wordLen = word.length();
    List<String> result = new ArrayList<String>();
    Queue<AbbreviatedWord> queue = new LinkedList<>();
    queue.add(new AbbreviatedWord(new StringBuilder(), 0, 0));
```

You must run your code first

## Time Complexity

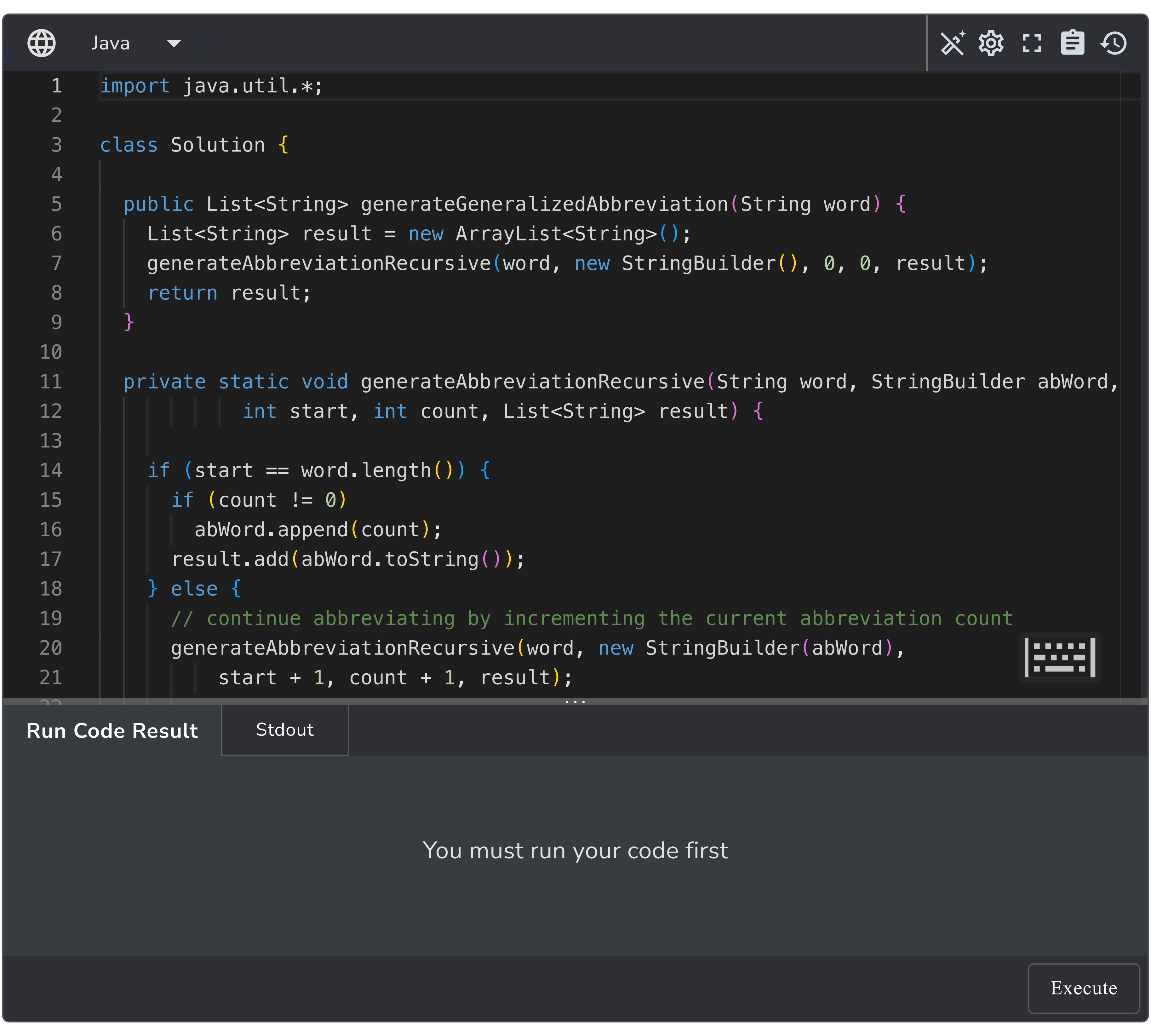
Since we had two options for each character, we will have a maximum of $2^N$ combinations. If you see the visual representation of Example-1 closely, you will realize that it is equivalent to a binary tree, where each node has two children. This means that we will have $2^N$ leaf nodes and $2^N - 1$ intermediate nodes, so the total number of elements pushed to the queue will be $2^N + 2^N - 1$, which is asymptotically equivalent to $O(2^N)$. While processing each element, we do need to concatenate the current string with a character. This operation will take $O(N)$, so the overall time complexity of our algorithm will be $O(N * 2^N)$.

## Space Complexity

All the additional space used by our algorithm is for the output list. Since we can’t have more than $O(2^N)$ combinations, the space complexity of our algorithm is $O(N * 2^N)$.

## Recursive Solution

Here is the recursive algorithm following a similar approach:



```java
import java.util.*;

class Solution {

  public List<String> generateGeneralizedAbbreviation(String word) {
    List<String> result = new ArrayList<String>();
    generateAbbreviationRecursive(word, new StringBuilder(), 0, 0, result);
    return result;
  }

  private static void generateAbbreviationRecursive(String word, StringBuilder abWord,
          int start, int count, List<String> result) {

    if (start == word.length()) {
      if (count != 0)
        abWord.append(count);
      result.add(abWord.toString());
    } else {
      // continue abbreviating by incrementing the current abbreviation count
      generateAbbreviationRecursive(word, new StringBuilder(abWord),
        start + 1, count + 1, result);
```

You must run your code first

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